Panel Method for NACA Airfoils

APPLIED AERODYNAMICS | PROJECT I

Inviscid, incompressible potential flow solver written in Python

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Scan to Open the GitHub Repo and Follow along

https://github.com/Malik-Saif-Ullah/panel_method_for_naca_airfoils

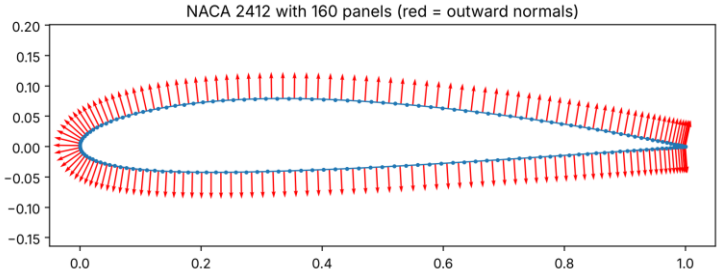
Problem and approach

Goal: pressure, lift, drag and the flow field of any 2D airfoil, ignoring viscosity.

1 **Discretise**
Cut the airfoil into 160 straight panels

2 **Unknowns**
Source per panel + one shared vortex

3 **Solve**
N + 1 equations, one numpy linear solve



NACA 2412, 160 panels. Red arrows = outward normals

Potential flow is linear: add a free stream, sources and a vortex, then solve for their strengths.
Hess-Smith Panel Method

The three equations

1 **Wall condition**

$\mathbf{V \cdot n = 0}$ **at every panel midpoint**

N equations: no flow through the surface

2 **Kutta condition**

V_t **(first panel) = -** V_t **(last panel)**

1 equation: smooth exit at trailing edge

3 **Linear system**

$\mathbf{Mx = b}$; $\mathbf{x = [\ \sigma_1 , \dots , \sigma_N , \gamma\ ]}$

N + 1 unknowns, solved in one line

FROM SOLUTION TO RESULTS

Pressure coefficient

$$C_{p,i} = 1 - (V_{t,i} / V\infty)^2$$

Pressure force

$$F_k = -\sum C_p \cdot n_k \cdot l$$

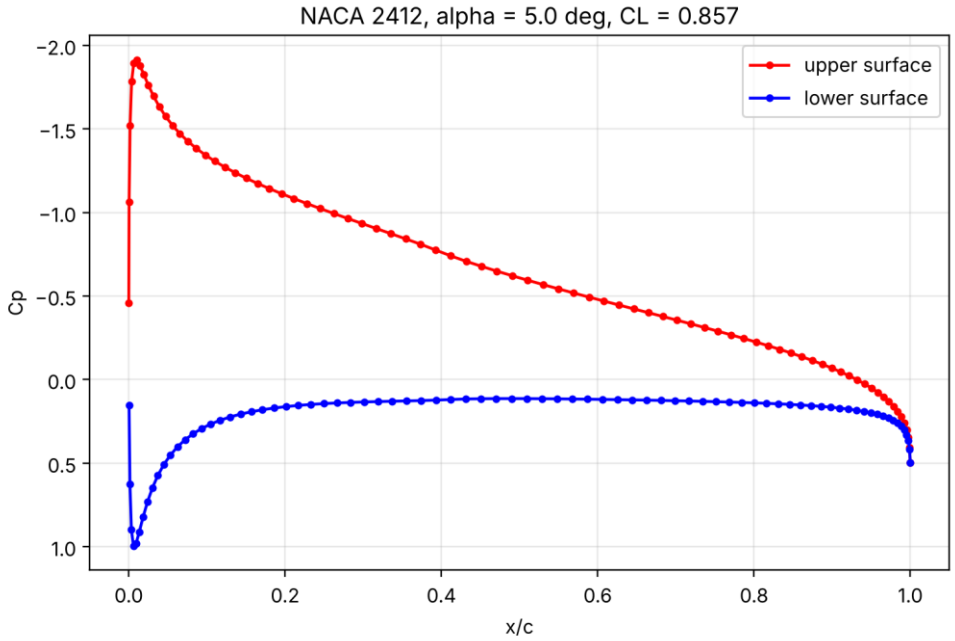
Lift

$$C_L = -F_x \sin\alpha + F_y \cos\alpha$$

Drag

$$C_D = F_x \cos\alpha + F_y \sin\alpha$$

Pressure coefficient (Cp)

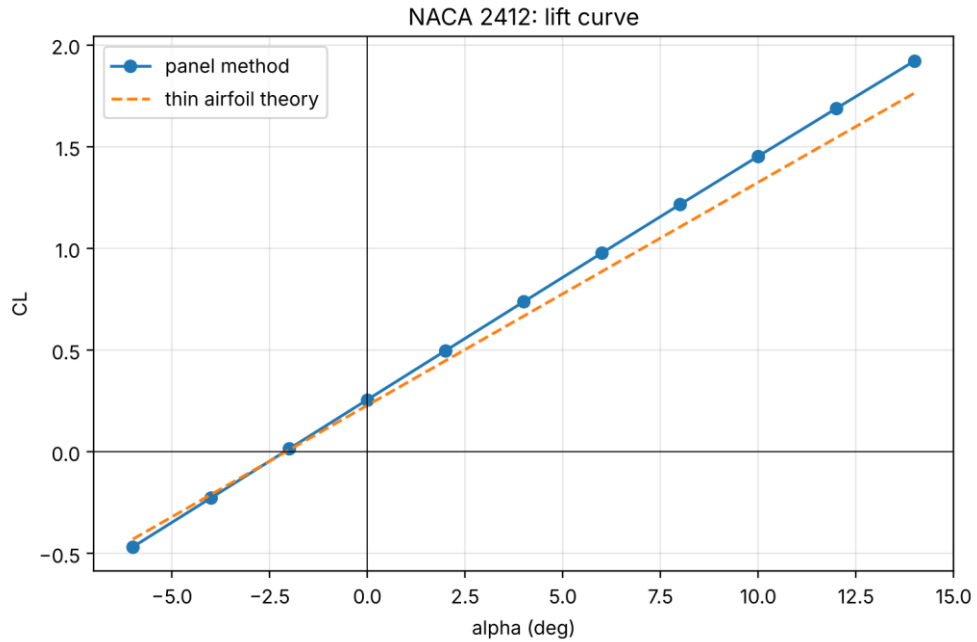


−1.91

suction peak at x/c = 0.010

- Upper surface: suction
- Lower surface: high pressure
- Area between curves = lift

Lift coefficient vs angle of attack

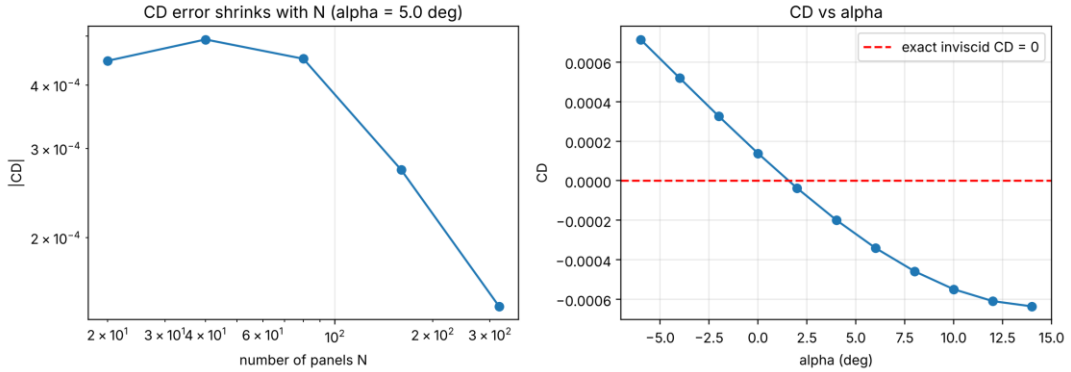


6.86 /rad
lift slope (theory 2π = 6.28)

−2.13°
zero-lift angle (theory −2.08°)

No stall in inviscid theory:
CL keeps rising

Drag coefficient (CD) analysis



−0.0003

panel method CD at N = 160 (ideal value 0)

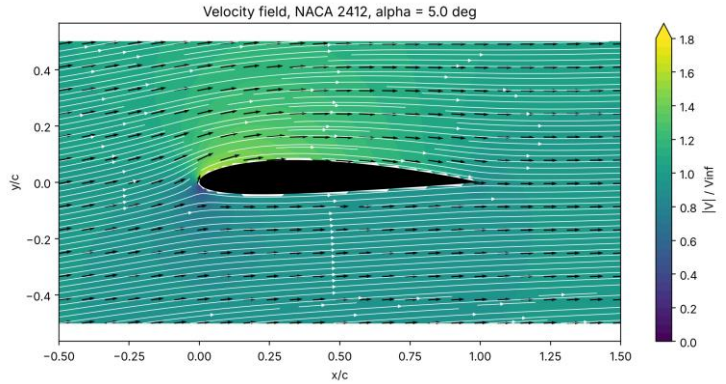
d'Alembert's paradox: ideal flow has zero drag. The tiny residual is numerical error and falls as N grows.

≈ 0.0099

rough viscous estimate, Re = 2×10^6

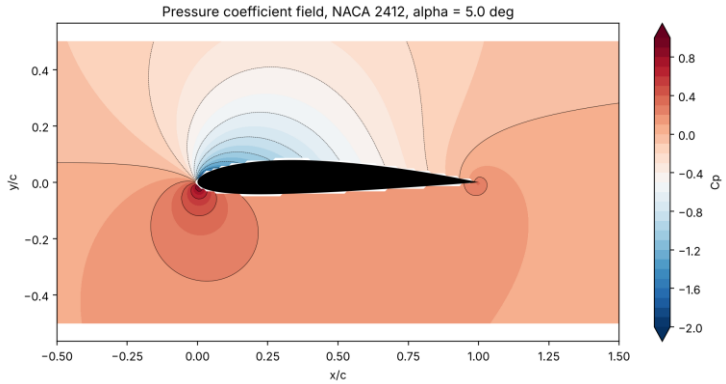
Velocity and pressure fields

Velocity field



Fast over the top (yellow), slow under the nose (blue). Streamlines bend around the airfoil.

Pressure field



Suction above (blue), high pressure under the nose (red). The difference is lift.

Validation and conclusions

Two routes, same lift

CL = 0.857 (pressure) and 0.857 (Kutta-Joukowski)

Matches thin airfoil theory

Slope 6.86 vs 6.28 per rad, zero-lift angle −2.13° vs −2.08°

Drag goes to zero

CD shrinks as panels increase (d'Alembert)

GOOD TO KNOW

- Inviscid: no stall, no real drag
- Incompressible: low Mach number only
- Any NACA 4-digit code: edit one cell and rerun

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Thank you!

Questions?